EXP: 4 Setting UP a Local Area Network (LAN)

Aim:

To Design, Set up and configure a LAN connecting four computers through a network switch, assign Static IP address to all devices, Verify connectivity using ping command, and demonstrate resource / data sharing across the network.

Equipment Required:

| S.NO | Equipment | Quantity | Specification |
|---|---|---|---|
| 1. | Personal computers | 4 | Ethernet enabled |
| 2. | Network Switch | 1 | 8 / 16 / 24 Port, unmanaged / managed |
| 3. | Ethernet Cables | 4 | Straight - through RJ - 45 connectors |

Implementation :

Step 1 : Place the Devices

* Open Cisco Packet Tracer
* From device category Panel, Select End Devices and drag 4 PCs onto workspace
* Select Switches and drag 1 switch onto the workspace

Step 2 : cable the Devices

* Click connections -> select copper Straight - Through cable
* Click each PC, then click the switch, and select an available FastEthernet Port complete each connection
* Repeat for all four PCs. Green link lights confirm successful Physical connection

Step 3 : Assign IP Addresses to PCs

* Click a PC -> Desktop tab -> IP configuration
* Enter the Static IP address and Subnet Mask as per the IP table for each PC.

Step 4 : Configure the Switch Management IP

* Click the Switch -> CLI tab
* Enter the following commands to assign management IP

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5 : Verify connectivity using Ping

* Click PC -> Desktop tab -> Command Prompt
* Use the Ping command to test connectivity to other PCs eg. Ping 10.1.1.2
* Successful Reply from ... messages confirm connectivity

Step 6 : Demonstrate Data Sharing

* Add Server device and connect it to switch
* Assign it IP address via its Desktop -> IP Configuration tab
* Open the Server Services tab and enable HTTP service
* From another PC, open Web Browser and enter server's IP address to load

Step 7: Observe Packet Flow in simulation mode

+ Switch to simulation mode
+ Send Ping or web request between two devices and use Play/capture controls to step through Packet it travels hop-by-hop through the switches observing MAC-address-based forwarding

Result:

All four PCs were successfully connected to the switch in a Star topology and assigned unique Static IP address within the same. Ping tests between all devices returned Successfully replies, confirming end-to-end connectivity. The shared folder/Server resource was successfully accessed from other computers networks-